Speech-to-Text API Documentation

Overview

The Speech-to-Text (STT) API provides automatic transcription services for audio and video files using the Whisper model, with support for Urdu language transcription.

Base URL: `https://stt-api-dgx.nimar.gov.pk`

API Key: `apikey1`

Authentication

The API uses API key authentication via a custom header.

Header:

```
X-API-Key: apikey1
```

Endpoints

1. Transcribe Video/Audio

Transcribe audio or video files to text with timestamps.

Endpoint: `POST /v1/transcribe`

Headers:

- `X-API-Key`: Your API key (required: `apikey1`)
- `X-Language`: Language code (required, e.g., `ur` for Urdu)

Request Body:

- Content-Type: `multipart/form-data`
- `video_file`: The audio or video file to transcribe (binary)

Response (200 OK):

```
json
```

```
{
  "full_text": "Complete transcribed text from the entire audio/video",
  "timestamp": [
    {
      "start": 0.0,
      "end": 5.5,
      "text": "First segment of transcription"
    },
    {
      "start": 5.5,
      "end": 12.3,
      "text": "Second segment of transcription"
    }
  ]
}
```

Response Fields:

- `full_text` (string|null): Complete transcription of the audio/video
- `timestamp` (array): Array of transcription segments
  - `start` (number): Start time in seconds
  - `end` (number): End time in seconds
  - `text` (string): Transcribed text for this segment

Error Response (422):

json

```
{
  "detail": [
    {
      "loc": ["body", "video_file"],
      "msg": "field required",
      "type": "value_error.missing"
    }
  ]
}
```

Example cURL:

bash

```bash
curl -X POST "https://stt-api-dgx.nimar.gov.pk/v1/transcribe" \
  -H "X-API-Key: apikey1" \
  -H "X-Language: ur" \
  -F "video_file=@/path/to/your/audio.mp3"
```

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2. Health Check

Check the health status of the API and Whisper server.

Endpoint: `GET /health`

Headers: None required

Response (200 OK):

json

```json
{
  "status": "healthy",
  "service": "Speech-to-Text API",
  "triton_server": "connected",
  "model_status": "ready"
}
```

Response Fields:

- `status` (string): Overall health status
- `service` (string): Service name
- `triton_server` (string): Triton server connection status
- `model_status` (string): Model availability status

Example cURL:

bash

```bash
curl "https://stt-api-dgx.nimar.gov.pk/health"
```

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3. Get Configuration

Retrieve current API configuration including supported languages.

Endpoint: `GET /config`

Headers: None required

Response (200 OK): Returns configuration object (check response for supported languages list)

Example cURL:

bash

```bash
curl "https://stt-api-dgx.nimar.gov.pk/config"
```

Note: Run this endpoint to get the list of supported languages before making transcription requests.

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4. API Information

Get basic API information.

Endpoint: `GET /`

Headers: None required

Response (200 OK): Returns API information and welcome message

Example cURL:

bash

```bash
curl "https://stt-api-dgx.nimar.gov.pk/"
```

Supported Languages

To get the full list of supported languages, call the `/config` endpoint. The API is confirmed to support:

- **Urdu** (`ur`)

Quick Start Guide

Step 1: Check API Health

bash

```bash
curl "https://stt-api-dgx.nimar.gov.pk/health"
```

Step 2: Get Supported Languages

bash

```bash
curl "https://stt-api-dgx.nimar.gov.pk/config"
```

Step 3: Transcribe Your File

bash

```bash
curl -X POST "https://stt-api-dgx.nimar.gov.pk/v1/transcribe" \
  -H "X-API-Key: apikey1" \
  -H "X-Language: ur" \
  -F "video_file=@your_audio_file.mp3"
```

Error Handling

Common Error Codes

Status Code	Description
200	Success
422	Validation Error (missing required fields or invalid input)
401	Unauthorized (invalid API key)
500	Internal Server Error

Validation Errors

The API returns detailed validation errors when requests are malformed:

json

```
{
 "detail": [
   {
    "loc": ["header", "X-Language"],
    "msg": "field required",
    "type": "value_error.missing"
   }
 ]
}
```

Best Practices

1. **Always check health before heavy usage** - Verify the API is healthy before processing multiple files
2. **Verify language support** - Call `/config` to confirm your target language is supported
3. **Handle errors gracefully** - Implement retry logic for transient failures
4. **Use appropriate file formats** - Common audio formats (MP3, WAV, M4A) and video formats are supported
5. **Monitor response times** - Large files may take longer to process

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Rate Limits

Contact your API administrator for specific rate limit information.

Support

For technical support or questions about the API, contact your development team lead or API administrator.